Taha Gamal Alieldin Mechatronics Engineering Graduate

tahajralieldin@gmail.com | 01032949396 | Mokattam, Cairo | linkedin.com/in/taha-alieldin

Military Status: Exempted | tahajralieldin-ship-it.github.io/taha-alieldin-portfolio

Summary

Mechatronics Engineering graduate specializing in automation, robotics, and machine learning. Led an 8-member team building an autonomous surveying swarm. Gained industrial experience at ENPPI and Norpetco, with strong communication and problem-solving skills from customer-facing roles.

Professional Experience

ENPPI, *Instrumentation & Control Intern* — 07/2025
Studied instrumentation and process control (sensors, transducers, P&ID) with exposure to SCADA, PLC, and DCS systems.

Norpetco (North Petroleum Company), *Production & Field Operations Trainee* — 07/2023 – 08/2023
Completed office and on-site field training, observing power generation, artificial lift equipment (Lufkin beam pumps, ESP), and firefighting systems. — Cairo (office) + Fields

Luban Workshop, Ain Shams University, *Embedded Robotics Trainee* — 08/2025
Built and programmed a differential-drive MicroMouse robot.

Sales & Receptionist, *Fitness Time Gym* — 2022 – 2025
#1 Sales Representative, generating ~300K EGP in monthly revenue. — Mokattam,Cairo

Certifications & Courses

Advanced Automation, *HA Consulting Group* — 06/2026 – Present
Expected Sep 2026.

Basic Automation, *HA Consulting Group* — 07/2024 – 09/2024
A+ in all three modules: Classic Control, PLC Basic Programming (ladder logic, Siemens TIA Portal), and Electric Motor & Drive Programming (VFDs).

OSHA Construction & Industry Safety Course, *Orascom Construction* — 02/2025

Education

Ain Shams University, *BSc. Mechatronics Engineering* — 09/2021 – 07/2026
- **GPA:** 2.75 (B-)
- **Graduation project grade:** A+

Selected Projects

AI-Enhanced Autonomous Surveying Swarm, *Graduation Project* — 08/2025 – 06/2026
- Led an 8-member team to build an autonomous surveying swarm on ROS 2 (Humble). Built Voronoi coverage algorithms and Gazebo simulations; pitched the prototype to win a 100K EGP grant from ASRT.

Real-Time Human Behavior Recognition System, *Hybrid CNN-LSTM Pipeline* — 10/2025 – 01/2026
- Directed a 4-member team building a real-time CNN-LSTM computer-vision system for action recognition, reaching the cohort's highest accuracy.

Furuta Pendulum Control System, — 10/2025 – 01/2026
LQR & Hardware-in-the-Loop Implementation via MATLAB/Simulink
- Designed an LQR controller to balance a rotary pendulum against disturbances, modeled in SimMechanics and validated via HIL, with a state-flow controller.

Technical Skills

- **Control & Automation:** PLC (ladder logic), classic control, LQR, state machines.
- **Programming & Tools:** MATLAB/Simulink, Python, C++, ROS 2, Gazebo, OpenCV, Automation Studio.
- **ML/Vision:** CNN, LSTM, deep learning.
- **Mechanical/Design:** Inventor, Ansys.